

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 1: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. (BL-6)

Assessment Tool Weightage: 40%

QUESTIONS:

Total Marks:25

Q1. Transfer Learning

A healthcare startup aims to develop an AI-based system for detecting skin cancer from dermoscopic images. Since only a limited number of labeled images are available, design a transfer learning strategy using a pre-trained CNN model. Justify your choice of model, identify which layers should be frozen or fine-tuned, and explain how your design improves classification performance.

Q2. DenseNet and PixelNet

A smart city project requires an automated road-scene analysis system that accurately identifies roads, pedestrians, vehicles, and traffic signs at the pixel level. Design a deep learning architecture using DenseNet and PixelNet to solve this problem. Explain how the proposed architecture improves feature reuse, segmentation accuracy, and computational efficiency.

Q3. Bidirectional RNN and Encoder–Decoder

A multinational customer service company wants to build a multilingual chatbot capable of translating customer queries while preserving contextual meaning. Design an Encoder–Decoder sequence-to-sequence model using Bidirectional RNNs. Illustrate the architecture and justify how bidirectional processing improves translation accuracy.

Q4. BPTT and LSTM

A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

Q5. Integrated Deep Learning Solution

A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for

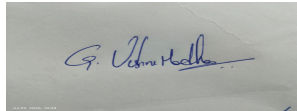
caption generation. Explain the role of each component and justify how the complete system produces accurate captions.

Code of Conduct:

I G Vishnu Madhav (192425192) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

1. INTRODUCTION

Deep Learning is a branch of Artificial Intelligence that uses multi-layer neural networks to learn complex patterns from data. It is widely used in healthcare, smart cities, finance, language translation, and multimedia applications.

Different deep learning architectures are designed for different types of problems. CNNs are mainly used for image processing, while RNNs and LSTMs are suitable for sequential and time-series data. Transfer Learning allows pre-trained models to be reused for new applications, especially when the available dataset is small.

This report presents five industry-oriented applications of deep learning: skin cancer detection, road-scene segmentation, multilingual translation, stock-price forecasting, and automatic video caption generation.

2. OBJECTIVES

The main objectives of this assessment are:

- To understand the application of deep learning in real-world industries.
- To design suitable deep learning architectures for different problems.
- To understand Transfer Learning and feature reuse.
- To apply RNN, LSTM, Encoder–Decoder, and BPTT concepts.
- To analyze the advantages and expected performance of the proposed solutions.

3. QUESTION 1 – TRANSFER LEARNING

Skin Cancer Detection Using a Pre-trained CNN

Problem

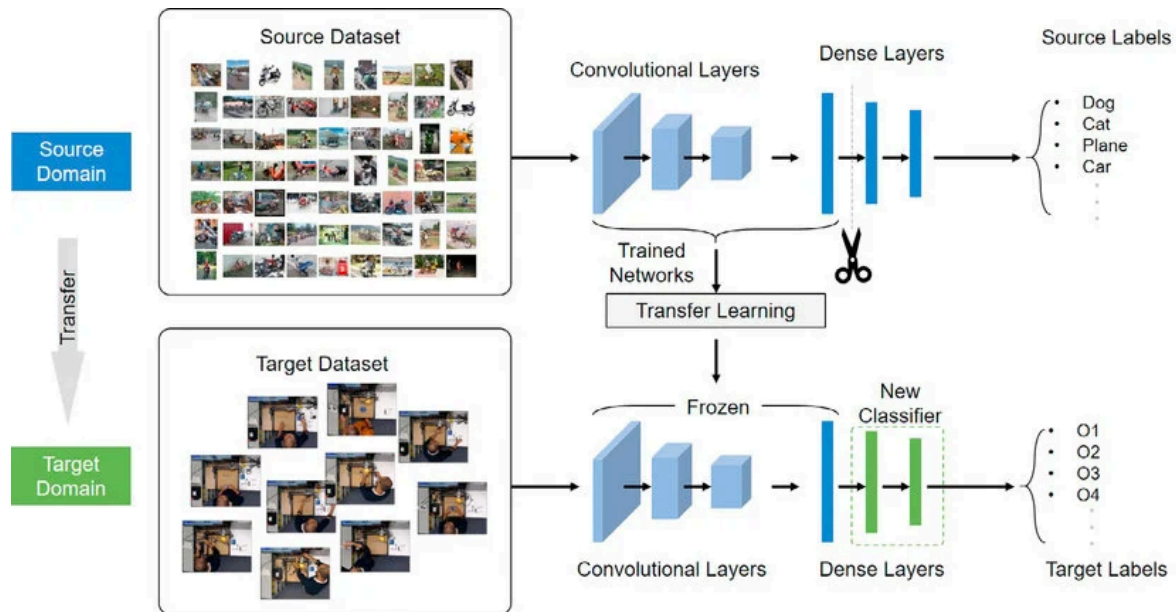
A healthcare startup wants to detect skin cancer from dermoscopic images. Since only a limited number of labeled images are available, training a deep CNN from scratch may result in poor performance.

Proposed Solution

A **ResNet50** pre-trained CNN can be used with Transfer Learning.

Architecture

Dermoscopic Image → Pre-processing → ResNet50 → Feature Extraction → Classification Layer → Benign/Malignant



Working

The pre-trained ResNet50 already contains useful visual features such as edges, textures, shapes, and patterns. The initial layers are frozen so that these general features are retained.

The final classification layer is replaced with a new layer suitable for skin cancer classification. After initial training, some of the deeper layers can be unfrozen and fine-tuned using a small learning rate.

Layer Strategy

- **Initial CNN layers:** Frozen
- **Deep convolutional layers:** Fine-tuned
- **New classification layer:** Fully trained
- **Output:** Benign or Malignant

Advantages

- Works effectively with limited training data.
- Reduces training time.
- Reuses previously learned features.
- Fine-tuning improves classification performance.

Expected Result

The system can provide automated skin-lesion classification and assist healthcare professionals in early screening.

4. QUESTION 2 – DENSENET AND PIXELNET

Road-Scene Analysis and Segmentation

Problem

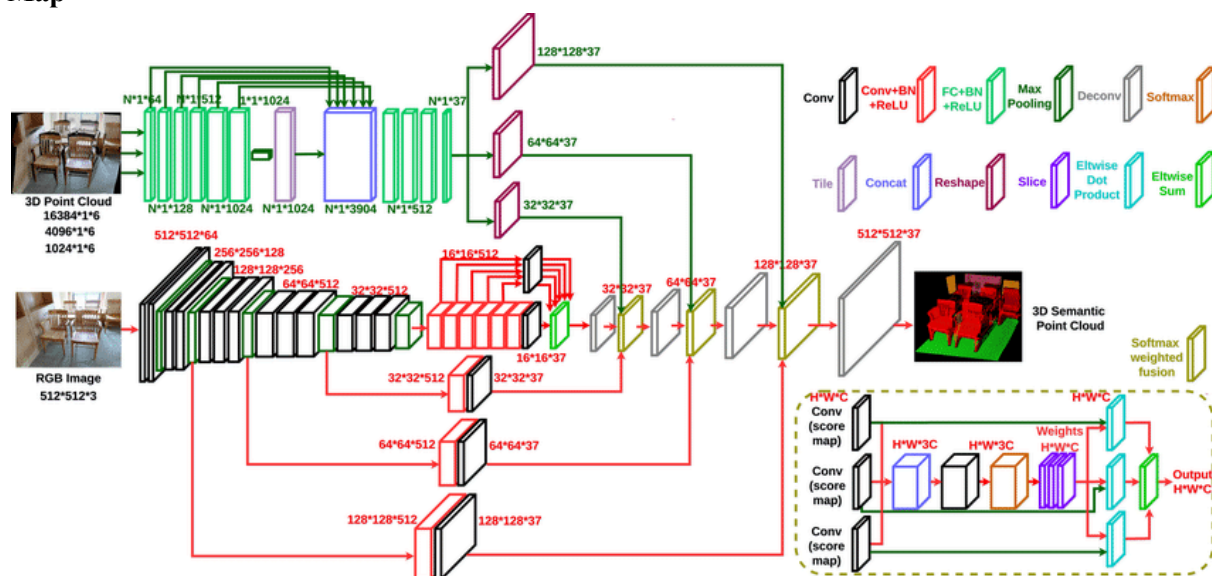
A smart city requires a system that identifies roads, pedestrians, vehicles, and traffic signs at the pixel level.

Proposed Solution

A DenseNet-based feature extractor can be combined with a PixelNet-style pixel-level prediction network.

Architecture

Road Image → DenseNet → Feature Maps → PixelNet → Pixel Classification → Segmentation Map



Working

DenseNet uses dense connections between layers. Therefore, features learned by earlier layers can be directly reused by later layers.

For example:

- Early layers → Edges and textures
- Middle layers → Shapes and patterns
- Deep layers → Vehicles, pedestrians and traffic signs

PixelNet performs pixel-level prediction, where every pixel is assigned to a particular class.

Feature Reuse

Dense connections allow information to flow directly between layers. This improves feature reuse and gradient flow while reducing unnecessary feature learning.

Segmentation

Each pixel can be classified as:

- Road
- Vehicle
- Pedestrian
- Traffic sign
- Background

Advantages

- Better feature reuse.
- Accurate pixel-level segmentation.
- Improved recognition of small objects.
- Efficient feature learning.

Expected Result

The system can generate a detailed road-scene segmentation map useful for intelligent traffic monitoring and autonomous transportation.

5. QUESTION 3 – BIDIRECTIONAL RNN AND ENCODER–DECODER

Multilingual Customer-Service Chatbot

Problem

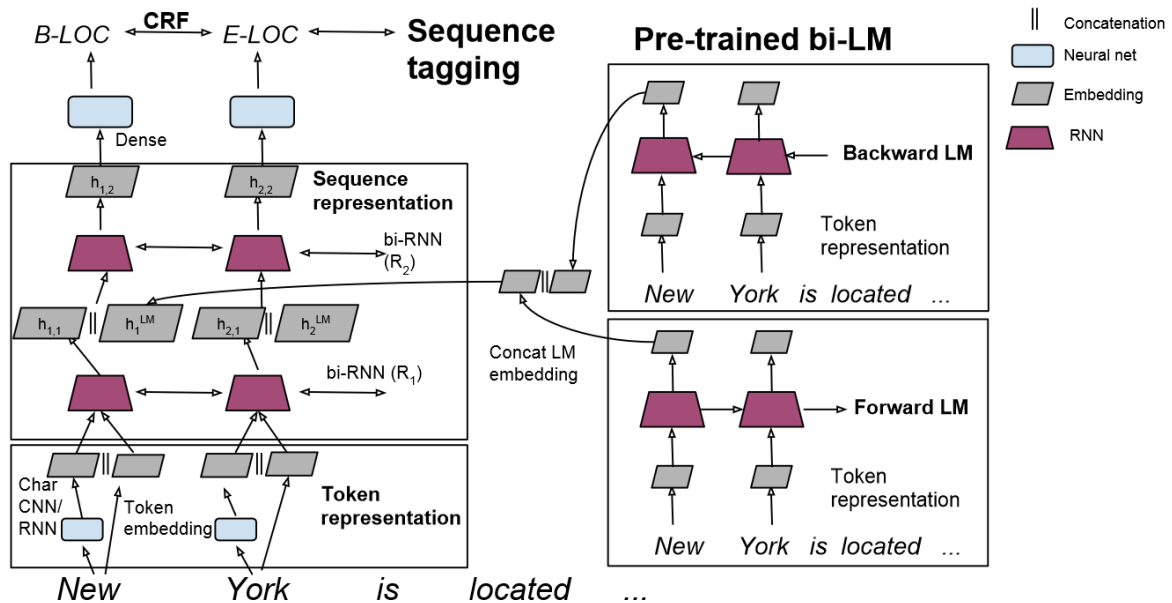
A multinational company wants a chatbot that can translate customer queries while preserving their contextual meaning.

Proposed Solution

A **Bidirectional RNN Encoder–Decoder** architecture is proposed.

Architecture

Input Sentence → Word Embedding → Bidirectional RNN Encoder → Context Representation → Decoder → Translated Sentence



Working

The input sentence is converted into word embeddings. The Bidirectional RNN processes the sentence in both directions.

Forward:

Word 1 → Word 2 → Word 3 → Word 4

Backward:

Word 4 → Word 3 → Word 2 → Word 1

The outputs from both directions provide information about the complete context of the sentence.

The decoder then uses this context to generate the translated sentence word by word.

Why Bidirectional RNN?

A normal RNN mainly considers information from the previous sequence. A Bidirectional RNN considers both previous and following words.

This helps the model understand:

- Sentence context
- Word relationships
- Grammar
- Meaning of ambiguous words

Expected Result

The proposed chatbot can provide more context-aware and accurate multilingual translations.

6. QUESTION 4 – BPTT AND LSTM

Stock-Price Forecasting

Problem

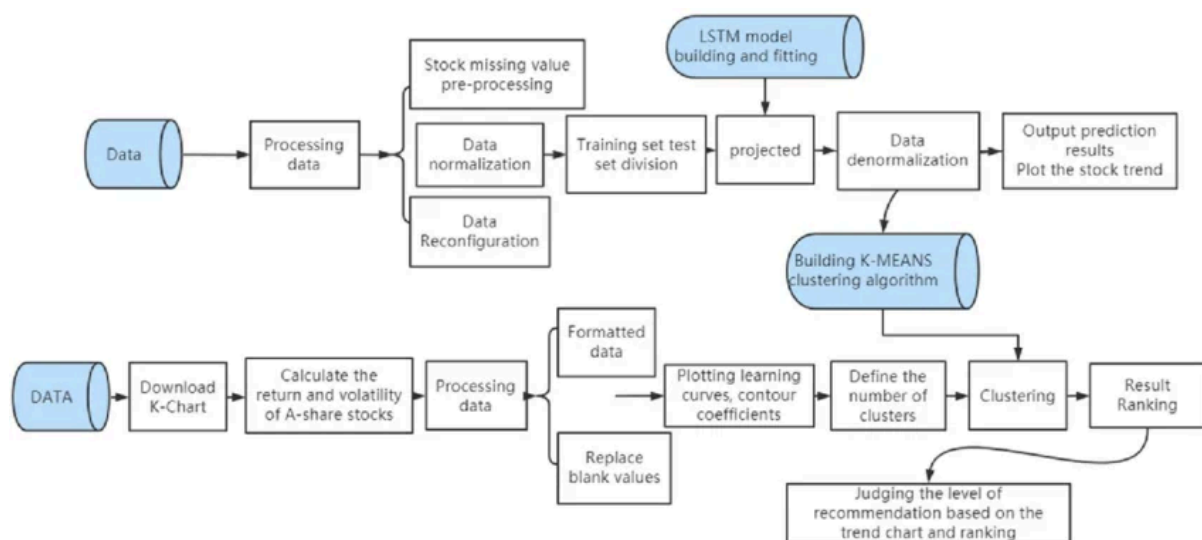
A financial analytics company wants to predict future stock prices using historical market data.

Proposed Solution

An **LSTM-based forecasting model** is proposed because stock prices are sequential and depend on historical patterns.

Architecture

Historical Market Data → Pre-processing → Sequence Creation → LSTM → Dense Layer → Predicted Price



Input Features

The model can use:

- Opening price
- Closing price
- High price
- Low price
- Trading volume

LSTM Working

LSTM contains three important gates:

- **Forget Gate:** Removes unnecessary information.

- **Input Gate:** Stores important new information.
- **Output Gate:** Produces the required information.

These mechanisms allow LSTM to remember important information over long sequences.

Role of BPTT

Backpropagation Through Time is used to train the LSTM.

The process is:

Input Sequence → Prediction → Loss Calculation → BPTT → Gradient Calculation → Weight Update

The error is propagated backward through multiple time steps to update the model parameters.

Why LSTM Instead of Traditional RNN?

Traditional RNNs can suffer from the vanishing-gradient problem and may forget long-term information. LSTM uses memory cells and gates to preserve important information for longer periods.

Expected Result

The model can learn historical market patterns and generate future stock-price predictions. Evaluation can be performed using **MAE, MSE, and RMSE**.

7. QUESTION 5 – INTEGRATED DEEP LEARNING SOLUTION

Automatic Video Caption Generation

Problem

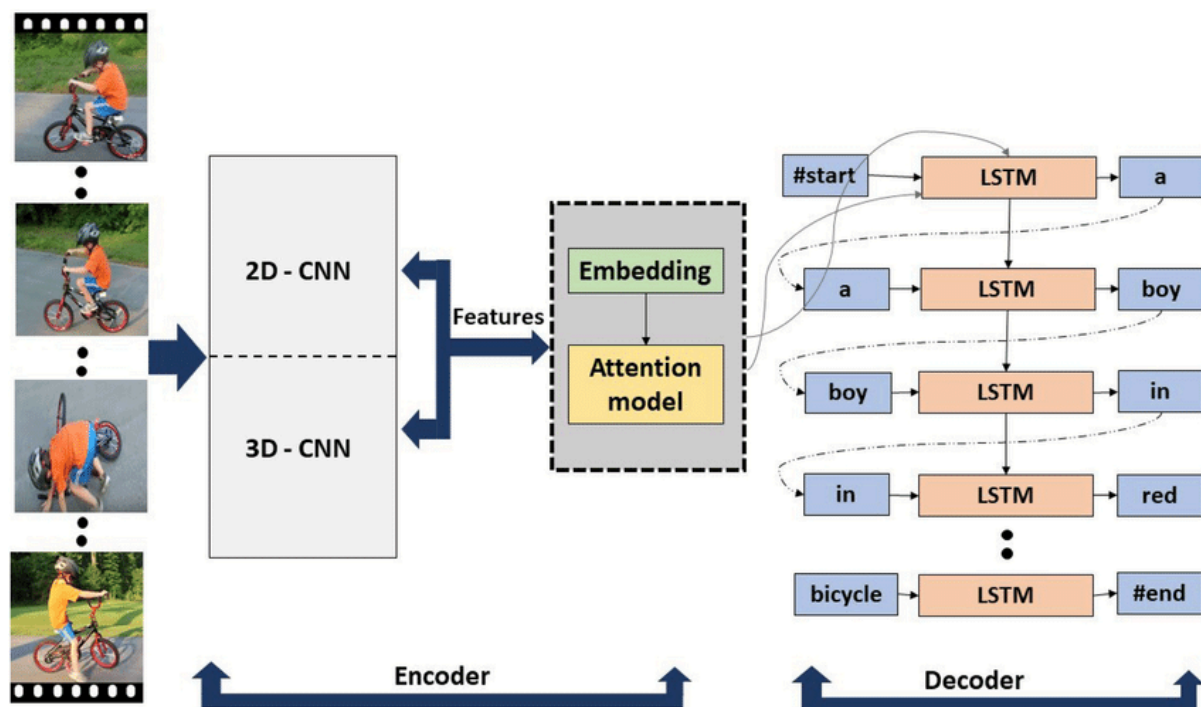
A media streaming company wants to automatically generate captions for uploaded videos to improve accessibility.

Proposed Solution

The system combines **Transfer Learning for visual feature extraction** with an **LSTM Encoder–Decoder for caption generation**.

Architecture

Video → Frame Extraction → Pre-trained CNN → Visual Features → LSTM Encoder → LSTM Decoder → Caption



Component 1 – Transfer Learning

A pre-trained CNN such as ResNet50 extracts visual information from individual video frames.

It identifies useful features such as:

- Objects
- People
- Scenes
- Actions

Instead of training the CNN from scratch, previously learned features are reused.

Component 2 – LSTM Encoder

The visual features from consecutive frames are given to the LSTM Encoder.

The encoder learns the temporal relationship between frames and creates a meaningful representation of the video.

Component 3 – LSTM Decoder

The decoder converts the encoded representation into natural-language text.

For example:

Video → Man + Running + Road → "A man is running on the road."

The caption is generated one word at a time.

Advantages

- Reuses pre-trained visual knowledge.
- Reduces training requirements.
- Understands temporal information.
- Generates captions automatically.
- Improves accessibility for users.

Expected Result

The complete system can analyze video content and generate meaningful captions automatically.

8. TOOLS AND TECHNOLOGIES

Category	Tools / Technologies
Programming	Python
Deep Learning	TensorFlow / Keras, PyTorch
CNN	ResNet50, DenseNet
Sequence Models	RNN, LSTM
Image Processing	OpenCV
Data Processing	NumPy, Pandas
Evaluation	Scikit-learn
Deployment	Flask / FastAPI
Hardware	GPU-enabled system / Cloud GPU

9. PERFORMANCE EVALUATION

Different metrics can be used according to the application.

Application	Evaluation Metrics
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Skin Cancer Detection	Accuracy, Precision, Recall, F1-score, AUC
Road Segmentation	IoU, Mean IoU, Pixel Accuracy
Translation	BLEU, ROUGE
Stock Prediction	MAE, MSE, RMSE
Video Captioning	BLEU, METEOR, ROUGE, CIDEr

These metrics help determine whether the proposed models provide reliable and accurate results

10. OVERALL CONCLUSION

This assessment demonstrates the use of different deep learning techniques for solving real-world industry problems.

Transfer Learning is useful when limited labeled data is available. **DenseNet and PixelNet** provide effective feature reuse and pixel-level segmentation. **Bidirectional RNNs and Encoder–Decoder models** improve contextual understanding for language translation. **LSTM and BPTT** are suitable for learning long-term dependencies in financial time-series data. Finally, combining **Transfer Learning with an LSTM Encoder–Decoder** enables automatic video caption generation.

Thus, selecting the appropriate deep learning architecture according to the problem, data type, and industry requirements can improve system accuracy, efficiency, and practical usability

11. REFERENCES

Hosny, K. M., Kassem, M. A., and Foad, M. M., “Classification of skin lesions using transfer learning and augmentation with Alex-net,” *PLOS ONE*, vol. 14, no. 5, 2019, e0217293. DOI: [10.1371/journal.pone.0217293](https://doi.org/10.1371/journal.pone.0217293).

Jain, S., Singhania, U., and Tripathy, B., “Deep Learning-Based Transfer Learning for Classification of Skin Cancer,” *Sensors*, vol. 21, no. 23, 2021, 8142. DOI: [10.3390/s21238142](https://doi.org/10.3390/s21238142).

Singhal, A., Shukla, R., Kankar, P. K., et al., “Comparing the capabilities of transfer learning models to detect skin lesion in humans,” *Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine*, vol. 234, no. 10, pp. 1083–1093, 2020. DOI: [10.1177/0954411920939829](https://doi.org/10.1177/0954411920939829).

Huang, G., Liu, Z., Van Der Maaten, L., and Weinberger, K. Q., “Densely Connected Convolutional Networks,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2261–2269, 2017. DOI: [10.1109/CVPR.2017.243](https://doi.org/10.1109/CVPR.2017.243).

